

COURSE NUMBER: CP3510

COURSE TITLE: Relational Database Design

COURSE DESCRIPTION:

This course introduces concepts common to all database management systems in such a way that the student can function in a meaningful and knowledgeable manner in any data processing environment where database concepts are implemented. The theoretical concepts are put into practice using current database architectures and technology.

PREREQUISITES: None

CO-REQUISITES: None

CREDIT VALUE: Three (3)

COURSE HOURS PER WEEK: Two (2)

LAB HOURS PER WEEK: Two (2)

REQUIRED TEXT:

Coronel, C., Morris, S., and Rob, P. (2019). *Database systems: Design, implementation, and management* (13th ed.). Toronto: Course Technology. eText ISBN: 9781337670562

LEARNING RESOURCES: None

MAJOR TOPICS:

- 1.0 Database Systems
- 2.0 Data Models
- 3.0 The Relational Database Model
- 4.0 Entity Relationship (E-R) Modeling
- 5.0 Normalization of Database Tables
- 6.0 Databases in Electronic Commerce
- 7.0 Using Databases

LEARNING OBJECTIVES:

1.0 Database Systems

- 1.1 Explain the difference(s) between data and information

- 1.2 Define the term database
- 1.3 Define the term database management system (DBMS)
- 1.4 Explain the relationship between a database and a database management system
- 1.5 List reasons why database design is important
- 1.6 Explain the historical roots of files and file systems
- 1.7 Identify problems with file system data management and explain how these issues can be resolved using a DBMS

2.0 Data Models

- 2.1 State the importance of data models
- 2.2 Identify the basic building blocks of data models
- 2.3 Explain the purpose of business rules in designing a database
- 2.4 Explain the evolution of data models
- 2.5 List the degrees of data abstraction

3.0 The Relational Database Model

- 3.1 Describe a logical view of data
- 3.2 Give definitions of the following types of keys:
 - 3.2.1 Super
 - 3.2.2 Primary
 - 3.2.3 Candidate
 - 3.2.4 Foreign
 - 3.2.5 Secondary
- 3.3 Define integrity rules
- 3.4 Apply the relational set operators
- 3.5 State the function of the data dictionary and the system catalogue
- 3.6 Define the term relationship as it applies to relational database
- 3.7 Explain the meaning of the term data redundancy
- 3.8 Define index
- 3.9 List Codd's relational database rules

4.0 Entity Relationship (E-R) Modeling

- 4.1 Describe the Entity-Relationship (E-R) model
- 4.2 Create an ER diagram using a diagramming tool
- 4.3 List database design challenges and conflicting goals

5.0 Normalization of Database Tables

- 5.1 Define the terms table and normalization as they apply to databases
- 5.2 Define the term functional dependency
- 5.3 Recognize functional dependencies in information

- 5.4 Convert a table with multivalued attributes into First Normal Form (1NF) using dependency diagrams
- 5.5 Convert a table in 1NF into Second Normal Form (2NF) using dependency diagrams
- 5.6 Convert a table in 2NF into Third Normal Form (3NF) using dependency diagrams
- 5.7 Convert a table in 3NF into Boyce-Codd Normal Form (4NF) using dependency diagrams
- 5.8 Improve a database design by:
 - 5.8.1 Evaluating primary assignment through surrogate (auto-number) keys
 - 5.8.2 Evaluating naming conventions
 - 5.8.3 Refining attribute atomicity
 - 5.8.4 Determining if there are new attributes to be added to a table
 - 5.8.5 Determining if there are new relationships to be added to a table
 - 5.8.6 Refining primary keys as required for data granularity
 - 5.8.7 Maintaining historical accuracy
 - 5.8.8 Evaluating derived attributes
- 5.9 Use unique indexes with surrogate keys to ensure data accuracy
- 5.10 Analyze higher-level Normal Forms
 - 5.10.1 Define the Fourth Normal Form (4NF)
 - 5.10.2 Explain normalization and database design
 - 5.10.3 Define denormalization

6.0 Databases in Electronic Commerce

- 6.1 Define Electronic Commerce
- 6.2 Explain the impact of E-Commerce
- 6.3 List E-Commerce styles
- 6.4 Define E-Commerce architecture
- 6.5 Explain security features required for E-Commerce
- 6.6 Explain web payment processing
- 6.7 Explain database design for E-Commerce applications
- 6.8 Implement E-Commerce databases

7.0 Using Databases

- 7.1 Create a table using CREATE TABLE command
- 7.2 Create single and multi-column primary key constraints
- 7.3 Create foreign key constraints
- 7.4 Use SQL statements to drop tables
- 7.5 Discuss SQL Integrity Constraints
- 7.6 Write SQL statements using integrity constraints including:
 - 7.6.1 UNIQUE constraints
 - 7.6.2 CHECK constraints using relational and logical operators
 - 7.6.3 NOT NULL constraints

- 7.7 Use SQL statements to:
 - 7.7.1 Add columns to tables using the ALTER TABLE command
 - 7.7.2 Modify tables using the ALTER TABLE command
 - 7.7.3 Delete columns from tables using the ALTER TABLE command
- 7.8 Enter data into tables using the SQL INSERT command
- 7.9 Modify data in tables using the SQL UPDATE and DELETE commands
- 7.10 Delete data in tables using the SQL DELETE commands
- 7.11 Create transactions using SQL statements
- 7.12 Create a transaction which uses the ROLLBACK, and COMMIT commands
- 7.13 Write SQL statements to select only certain rows from a table using the WHERE clause
- 7.14 Create SQL code using the IN and BETWEEN ... AND relational operators
- 7.15 Select only certain rows from a table using the WHERE clause with multiple conditions
- 7.16 Test for NULL values
- 7.17 Join tables using the traditional method of joining tables using the WHERE clause
- 7.18 Use table and column aliases

EVALUATION:

Assignments	30%
Test(s)	30%
Final	40%

DATE DEVELOPED: February 2010

DATE REVIEWED:

REVISION NUMBER: 2

DATE REVISED: September 2020

Effective PeopleSoft Date: September 2021

Note to instructor: Check PIRS to ensure this outline is the most current version.